DOD-V-15218C METRIC
21 November 1977
SUPERSEDING
MIL-V-15218B
1 November 1967
(See 6.7)

MILITARY SPECIFICATION

VARNISH (MIXING, PHENOLIC) (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two phenolic mixing varnishes for use in mixing with alkyd resin in the preparation of paint. Both varnishes are 25 gallons in oil length (i.e., for every 100 pounds of resin, 25 gallons of oil are used).

1.2 Classification. Varnish shall be of the following types, as specified (see 6.2):

Type I - Para-phenyl phenolic resin; 50:50 tung and linseed oils.
Type II - Alkyl phenolic resin; 50:50 tung and linseed oils.

APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-L-1155 - Linseed Oil, Alkali Refined.
TT-R-266 - Resin, Alkyd; Solutions.
TT-R-271 - Resin, Phenol-formaldehyde (Para-phenyl and Para-butyl).
TT-T-291 - Thinner-Paint, Volatile Spirits, Petroleum Spirits.
TT-T-775 - Tung Oil, Raw (China Wood) (For Use In Organic Coatings).
TT-X-916 - Xylene (For Use In Organic Coatings).
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY
MIL-R-15189 - Resin, Phenolic (Alkyl Phenol, Oil-Soluble).

STANDARDS

FEDERAL
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 56 - Flash Point by Tag Closed Tester, Test for.
D 600 - Liquid Paint Driers, Spec. for.
D 1296 - Odor of Volatile Solvents and Diluents, Test for.
D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes, Test for.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products, Test for.
D 1542 - Rosin in Varnishes, Qualitative Tests for.
D 1544 - Color of Transparent Liquids (Gardner Color Scale), Test for.
D 1545 - Viscosity of Transparent Liquids by Bubble Time Method, Test for.
D 1642 - Elasticity or Toughness of Varnishes, Tests for.
D 1643 - Gas Checking and Draft Test of Varnish Films, Test for.
D 1951 - Ash in Drying Oils and Fatty Acids, Test for.
D 2369 - Volatile Content of Paints, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.5). Questions pertinent to this effect shall be referred by the procuring activity to the appropriate service medical department which will act as advisor to the procuring activity.

3.2 Formulas. The varnish shall consist of ingredients conforming to the applicable specifications in the proportions shown in table I. The amount of thinner shall be adjusted to produce a varnish of the required nonvolatile content.

TABLE I Formulas.

Ingredients	Type I		Type II	
	kg (lbs.)	Gal.	kg (lbs.)	Gal.
Resin, para-phenyl-formaldehyde (TT-R-271)	45.5 (100)	10.1 1/	---	---
Resin, phenolic, alkyl phenyl oil-soluble (MIL-R-15189)	---	---	45.5 (100)	12.0 1/
Linseed oil, alkali refined (TT-L-1155)	44.1 (97)	12.5	44.1 (97)	12.5
Tung oil, raw (China wood) (TT-T-775)	44.5 (98)	12.5	44.5 (98)	12.5
Thinner-paint (type I or type II, grade A, TT-T-291)	73.2 (161)	24.5	---	---
Thinner-paint (type II, grade A, TT-T-291)	---	---	91.4 (201)	30.5
Xylene (grade B of TT-X-916)	19.5 (43)	6.0	---	---

1/ The volume given is for resin of minimum specific gravity. It will be slightly less when resin of maximum specific gravity is used.

3.3 Manufacture. The ingredients specified in table I shall be properly combined to produce a stable, uniform varnish in full conformity with the requirements of this specification.

3.4 Quantitative requirements. The varnish shall conform to the quantitative requirements shown in table II.

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TABLE II. Quantitative requirements.

Characteristics	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Nonvolatile content, percent by weight	59	61	59	61
Viscosity, Gardner[1/]	L	N	L	N
Ash, percent	--	0.05	--	0.05
Weight per gallon, kg (pounds)	3.45(7.6)	3.55(1.8)	3.41(7.5)	3.5(7.7)
Flash point, °C (°F)	30(86)	--	38(100)	--
Color, Gardner Color Standards	--	14	--	15
Drying time after thinners and driers added (see 4.3.1)				
set-to-touch, hours	--	2	--	2
dry hard, hours	--	8	--	12

[1/] Varnish incorporated into a paint and not procured directly by the Government as a varnish shall be considered as conforming in viscosity when it is no less than K and no more than O in viscosity (see 6.6).

3.5 Qualitative requirements. The varnish shall conform to the following qualitative requirements:

3.5.1 Appearance. The varnish shall be clear, uniform, and free from grain, dots, gel bodies, skins, and other insoluble matter that may be in suspension.

3.5.2 Odor. The odor shall be characteristic of the volatiles permitted when tested in accordance with 4.3.2.

3.5.3 Rosin and rosin derivatives. Rosin, rosin derivatives, and material and synthetic resins (other than phenolic resins conforming to TT-R-271 or MIL-R-15189) shall not be present when tested in accordance with 4.3.2.

3.5.4 Phenolic resins. Varnish, type I, shall test positive for the presence of para-phenyl-phenol-formaldehyde resin. Varnish, type II, shall test positive for the presence of phenol-formaldehyde resins when tested as specified (see 4.3.1 and 4.3.2).

3.5.5 Compatibility with alkyd resins. The varnish shall show no evidence of incompatibility when tested as specified (see 4.3.4).

3.5.6 Toughness. The varnish shall pass the following rosin pentaerythritol ester reduction when tested as specified (see 4.3.5):

Types	Rosin-pentaerythritol ester reduction percent
I	75
II	90

3.5.7 Immersion resistance. There shall be no evidence of permanent whitening, dulling, checking, blistering, or other signs of deterioration when tested as specified (see 4.3.6).

3.5.8 Alkali resistance. The varnish shall withstand the test specified (see 4.3.7). The film shall be intact and shall show no noticeable discoloration. A very slight hazing will be permitted if, after light buffing with the hand, no permanent hazing or marking remains.

3.5.9 Skinning. The varnish shall show no skinning at the end of 48 hours when tested as specified (see 4.3.2).

3.5.10 Gasproofness. The varnish films shall show no evidence of crow's-footing, wrinkling, checking, frosting, or any other defect when tested as specified (see 4.3.2).

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3.5.11 Volatile portion (type II). The volatile portion of type II varnish shall conconform to the following requirements by volume.

(a) A combination of hydrocarbons, alcohols, aldehydes, ethers, esters, or ketones having an olefinic or cycloolefinic type of unsaturation except perchloroethylene: 5 percent maximum.
(b) A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene, methyl benzoate, and phenyl acetate: 8 percent maximum.
(c) A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene, or toluene: 20 percent maximum.

3.5.12 Material Safety Data Sheet. The procuring activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD form 1813 and found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be provided in accordance with method 1031 of FED-STD-141.

4.3 Test procedures.

4.3.1 Tests for compliance with 3.4 (with exception of drying time) and 3.5.1 to 3.5.5, inclusive, shall be conducted on the material as received. Tests for compliance with 3.5.6 to 3.5.10, inclusive, shall be conducted after thinning the material as received to a Gardner viscosity of C-E, using the thinners as specified in the formula for each varnish (see 3.2) and after adding naphthenate driers conforming to TT-D-643. The driers shall be added a minimum of 48 hours before tests are conducted and in the following percentages of metals based on the amounts of oil present: 0.5 lead, 0.05 cobalt, and 0.02 manganese.

4.3.2 Tests shall be conducted in accordance with the applicable test methods shown in table III.

TABLE III. Test procedures.

Test	Applicable method in FED-STD-141	Applicable ASTM test method
Nonvolatile content	----	D 2369
Viscosity	----	D 1545
Ash	----	D 1951
Weight per gallon	----	D 1475
Flash point	----	D 56
Color	----	D 1544
Drying time	4061	----
Appearance	4261	----
Odor	----	D 1296
Rosin and rosin derivatives	----	D 1542
Phenolic resins	5141	----
Toughness (rosin-pentaerthritol ester reduction)	----	D 1642, method A
Immersion resistance	----	D 1308, section 5
Skinning	3021	----
Gasproofness	----	D 1643, method B

4.3.3 Drying time. Flow out the reduced varnish (see 4.3.1) on a properly toluene-cleaned glass panel; place the panel in a nearly vertical position in a well-ventillated room. Proceed in accordance with method 4061 of FED-STD-141.

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4.3.4 Compatibility with alkyd resin. Place a 50-milliliter (-mL) sample of the varnish in a 100-mL clean, dry, glass-stoppered, graduated cylinder. Add 50 mL of alkyd resin solution conforming to type I, class A of TT-R-266 and agitate until well mixed. Examine for cloudiness, precipitation, or other signs of incompatibility. After allowing to stand 24 hours, re-examine.

4.3.5 Toughness (rosin-pentaerythritol ester reduction.) Rosin-pentaerythritol ester reduction shall be determined as specified in method A of ASTM D 1642, except that the percent reduction by weight shall be made as specified (see 3.5.6).

4.3.6 Immersion resistance. Prepare a flow-out panel on tin and proceed according to ASTM D 1308. Allow the panel coated with the reduced varnish (see 4.3.1) to dry 72 hours prior to immersion. Immersion shall be in distilled water at 100°C (212°F) for a period of 7 hours. A recovery period of 2 hours shall be allowed before examination of the panel under test.

4.3.7 Alkali resistance. Dip the closed end of a well-cleaned 2.54-centimeters(-cm) by 15.24-cm (1-inch by 6-inch) test tube to a depth of 10.16 cm (4 inches) into the reduced varnish (see 4.3.1) and allow to dry in an inverted position, that is, with the mouth of the test tube down, for 48 hours. Place the end of the test tube which was uppermost during the drying period 5.18 cm (2 inches) below the surface of 5 percent sodium hydroxide at 20°C (68°F) for 7 hours. After removal, immediately rinse under gently running tap water, holding near the mouth of the tube and being careful not to touch the treated film. Dry for 2 hours and examine under suitable light for dulling, checking, or blistering. Defects at the extreme bottom of the test tube shall be disregarded.

4.4 Inspection of preparation for delivery. The packaging, packing, and marking of the varnish shall be inspected to determine compliance with section 5.

4.5 Toxicity. A manufacturer of material shall disclose the formulation of his product to the Navy Bureau of Medicine and Surgery, Navy Department, Washington, DC 20372. The disclosure of proprietary information, which shall be held in confidence by the Bureau of Medicine and Surgery, shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, or disposal of the material.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging, packing, and marking. The varnish shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be A or C and the level of packing shall be A, B, or C as specified (see 6.2).

5.1.1 Special marking. In addition to other markings required on the containers, there shall be the following statement:

"The volatile content of this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District." (See 6.5.)

6. NOTES

6.1 Intended use. This specification covers an ingredient for use in specifications for paints and related materials and, as such, is not intended for use for direct procurement by the Government. In the event of small purchases from industry, the special marking requirements specified herein (see 5.1.1) shall apply. The Quality Assurance Provisions are provided for use in connection with this material when used as an ingredient in paints purchased by the Government.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Levels of packaging and packing required (see 5.1).
(d) Special marking required (see 5.1.1).

6.3 Types III and IV of MIL-V-15218B varnish have been eliminated. Type III is no longer used and type IV is an alternate choice to type II.

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6.4 Cooking schedules. The following schedules have been used successfully to cook the varnish:

6.4.1 Type I. Heat one-half of the resin and all of the linseed oil to 293°C (560°F) in 30 minutes. Hold 1 hour at 293°C (560°F) and then add the remaining resin. After this is melted, add the tung oil and reheat to 241°C (465°F). Hold for proper bodying. Cool and thin. No driers shall be added to the clear varnish.

6.4.2 Type II. Heat one-half of the resin and all of the linseed oil to 293°C (560°F) in 30 minutes. Hold 1-1/2 hours at 293°C (560°F) and then add the remaining resin. After this is melted, add the tung oil and reheat to 274°C (525°F). Hold for proper bodying. Cool and thin. No driers shall be added to the clear varnish.

6.5 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District, the paint may be used anywhere else paint complying with 3.5 is allowed. This includes other air pollution control districts or similar areas controlling this emission of solvents into the atmosphere. Information regarding Los Angeles County Air Pollution Rules 102, 442, and 443 may be obtained from: South Coast Air Quality Management District, Metropolitan Zone, 434 South Pedro Street, Los Angeles, California 90013.

6.6 Varnish in paint. The viscosity limits contained in table II specify a standardized product suitable for use in any of a number of mandatory formula-type paints. However, on occasion, paint manufacturers may desire a slightly higher or lower viscosity varnish for specific batches of paint in order to compensate for the allowable variations in other ingredients incorporated in that paint. In such cases, it is permissible for the paint manufacturer to use varnish conforming to this specification and whose viscosity may include K or O as indicated in the footnote to table II. It is the responsibility of the paint manufacturer to select a varnish within this viscosity range that results in a paint that conforms in all respects to the paint specification.

6.7 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous isse, due to the extensiveness of the changes.

Custodians:
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